

FIM 590 Algo. and HFT

Homework #1

Due: September 10th, 2026

Let W_t be a standard Brownian motion.

1. Write an HJB equation for $\inf_{\pi} \mathbb{E} \left[\frac{\gamma}{2} \int_0^T \pi_s^2 + X_T^2 \right]$ where $dX_t = -\pi_t dt + \sigma dW_t$ with $X_0 = x \neq 0$, and $\gamma > 0$. Solve the HJB explicitly using ansatz $H(t, x) = a(t)x^2 + b(t)$.
2. Consider a risky asset price

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t .$$

For a portfolio with self-financing condition

$$dX_t^\pi = X_t^\pi \pi_t \cdot \frac{dS_t}{S_t} + r(1 - \pi_t)X_t^\pi dt - c_t dt ,$$

where π_t is the portfolio's weight in the risky asset, $c_t \geq 0$ is rate of consumption, and r is the risk-free rate. Find the optimal consumption and the optimal value for

$$\sup_{\pi} \mathbb{E} \int_0^\infty e^{-\rho t} \log(c_t) dt ,$$

where $\rho > 0$ is the utility discount factor.

3. Given that the summed squared increments of W_t over $[0, T]$ converge to T with probability one, show that Brownian motion has infinite total variation with probability one.

Questions? Please write to apapani@ncsu.edu.